Belief Geometry on the Random Hierarchy Model

A small transformer partially encodes — and causally uses — the exact tree posterior, probed against ground truth

simplex-rhm-belief · run of 29 June 2026 · all numbers from the recorded pipeline output

Abstract. We reproduce, in miniature, the Simplex belief-geometry result on a hierarchical generative process. We train a 2-layer GPT-2 decoder (≈399k parameters) by next-token prediction on samples from a Random Hierarchy Model (RHM; arity $s = 2$, depth $L = 3$, vocabulary $v = 8$), a grammar small enough (1024 equiprobable trees) to enumerate exactly. We compute the **exact** Bayesian posterior over the tree's hidden latents by sum-product belief propagation, verified against brute-force enumeration to $< 10^{-6}$. The trained model reaches a test cross-entropy of 0.88 nats, closing ≈89% of the gap between the uniform baseline (2.08) and the Bayes-optimal floor (0.73) — it has effectively learned the posterior predictor. A single global linear probe partially recovers this posterior from the residual stream: held-out $R^2 = 0.38$ for the root class versus ≈0 for a shuffled control — a real but noisy affine image, not an exact reconstruction — with decodability accumulating across depth ($-0.00 \to 0.15 \to 0.38$) and the belief readout **blooming** outward from the prior toward simplex vertices as context accumulates. A latent-level sweep shows the residual encodes the **whole** hierarchy, most strongly the locally-predictive deep latents (R^2 up to 0.66) and least strongly the coarse global root. Causal steering on the layer-1 residual confirms the belief is **used**, not merely decodable: steering toward a wrong latent collapses the true next token's log-probability ($-0.73 \to -8.5$). We pre-registered our predictions before seeing results; three of four held, and the one miss (root $R^2 < 0.5$) is informative. Finally, a 30-run sweep over 10 random grammars × 3 replicate seeds shows all four findings — the inverted level gradient, blooming, layer accumulation, and causal steering — replicate across the RHM family with low replicate variance, and that the published `grammar = 0` numbers sit inside the family distribution. A further 99-run architecture sweep (width, depth, heads, and training budget, one-axis-at-a-time × 3 grammars × 3 seeds) shows the phenomenology is capacity-robust down to a low-width floor: decodability rises with width and depth, the inverted gradient holds at every capacity, heads matter least, and decodability is only weakly correlated with how well the model fit the loss.

not holds

1. Introduction

Natural data is hierarchical: characters compose into words, words into phrases, phrases into meaning. The Random Hierarchy Model (RHM) of Cagnetta et al. abstracts this into a clean synthetic grammar — a fixed tree in which each high-level symbol expands, via random production rules, into a fixed-length string of lower-level symbols, down to observed leaves. Predicting the next leaf optimally requires inferring the distribution over the **hidden** latent symbols that generated the observed prefix. The optimal next-token predictor **is** the Bayesian belief state over those latents.

This makes the RHM an ideal testbed for **belief geometry**: the hypothesis, developed in the transformer-interpretability literature, that a network trained by next-token prediction comes to represent, in its residual stream, the posterior distribution over the data-generating process's hidden state — and that this distribution is laid out as a linear (affine) image of the probability simplex. If true, the belief should be (i) linearly decodable, (ii) geometrically organized as a simplex that sharpens with context, (iii) built up additively across layers, and (iv) causally relevant to the model's output.

The RHM is uniquely suited to test all four claims **exactly**. Unlike natural language, its hidden state has a precise definition and its posterior is computable in closed form by belief propagation on the tree. With small parameters the entire grammar is enumerable, so the probe target is ground truth — not an estimate. This paper trains a tiny transformer on such a grammar and asks whether the residual stream carries the exact posterior, where in the network it lives, and whether the model relies on it.

2. Experimental design

2.1. The Random Hierarchy Model and exact beliefs

Grammar. We use the defaults of the spec: arity $s = 2$, depth $L = 3$, so each string has $d = s^L = 8$ leaves; per-level vocabulary $v = 8$; and $m = 2$ production rules per parent symbol, chosen uniformly. Rules are drawn once and held fixed, and constrained to be **unambiguous** (the spec's quick-failure check confirms no two parents share a production). A sample is generated top-down: pick a root class uniformly, recursively expand each symbol by a uniformly chosen rule, and emit the leaf string. With these parameters the grammar admits exactly $v \cdot m^{d-1} = 8 \cdot 2^7 = 1024$ equiprobable distinct trees — small enough to enumerate completely.

The exact frozen grammar used in every result is shown below (Table 1). Symbols are written as `S1`, …, `S8` for readability (the saved arrays use zero-based ids); at each level the parent takes one of the two listed child pairs uniformly.

Parent	Top expansion	Middle expansion	Bottom expansion
`S1`	`[S3, S1]` or `[S4, S8]`	`[S2, S1]` or `[S1, S5]`	`[S4, S1]` or `[S5, S3]`
`S2`	`[S4, S3]` or `[S1, S3]`	`[S6, S3]` or `[S3, S1]`	`[S6, S7]` or `[S8, S8]`
`S3`	`[S6, S7]` or `[S5, S6]`	`[S5, S2]` or `[S1, S1]`	`[S3, S4]` or `[S7, S8]`
`S4`	`[S6, S8]` or `[S7, S7]`	`[S6, S5]` or `[S8, S6]`	`[S6, S5]` or `[S4, S7]`
`S5`	`[S6, S2]` or `[S1, S5]`	`[S4, S3]` or `[S1, S8]`	`[S1, S5]` or `[S6, S4]`
`S6`	`[S8, S6]` or `[S2, S7]`	`[S7, S6]` or `[S4, S5]`	`[S8, S4]` or `[S1, S8]`
`S7`	`[S4, S7]` or `[S1, S1]`	`[S3, S8]` or `[S4, S1]`	`[S4, S6]` or `[S3, S2]`
`S8`	`[S2, S2]` or `[S5, S7]`	`[S1, S2]` or `[S8, S3]`	`[S5, S7]` or `[S3, S6]`

Table 1: Frozen level-specific grammar sampled once before training. Top, middle, and bottom columns correspond to `rules_0`, `rules_1`, and `rules_2` in `artifacts/grammar.npz` (see visualized in appendix Appendix 9).

Exact posterior. Given a leaf prefix of length k, the posterior over any hidden latent (including the root class) is computed by sum-product belief propagation on the tree: upward messages from observed leaves combine through the production rules to give the marginal over each latent. Because the grammar is fully enumerable, we verified the belief-propagation posterior against brute-force enumeration of all consistent trees; the two agree to $< 10^{-6}$ across 10 passing tests, alongside checks that sample length equals s^L and that unambiguity holds. The recorded self-test (a representative string) shows the root posterior sharpening with context — entropy 1.89 nats at $k = 1$ falling to 0.00 (a single consistent root) by $k = 3$ — which previews the blooming geometry below.

2.2. Model and training

We train a HuggingFace `GPT2LMHeadModel` with 2 layers, `n_embd` = 128, 4 heads, and all dropout disabled, totalling 398,848 parameters. There is no tokenizer: RHM leaf symbols are integers fed directly as `input_ids`, with `vocab_size` = 8 and `n_positions` = 8. We train by next-token prediction on 922 training strings (102 held out) for 4000 steps on Apple MPS.

The Bayes-optimal floor. The value of an enumerable grammar is that we know the best achievable loss. Averaging the exact per-position posterior entropy over the seven predicted positions gives the Bayes-optimal mean next-token cross-entropy: 0.725 nats. The uniform baseline is $\ln 8 = 2.079$. The trained model reaches a final held-out cross-entropy of **0.880 nats** (Figure 1), closing 89% of the gap. The model has, to a good approximation, learned the true posterior predictor — which is the precondition for asking whether it represents the posterior internally. Later this is reproduced at scale (more grammars).

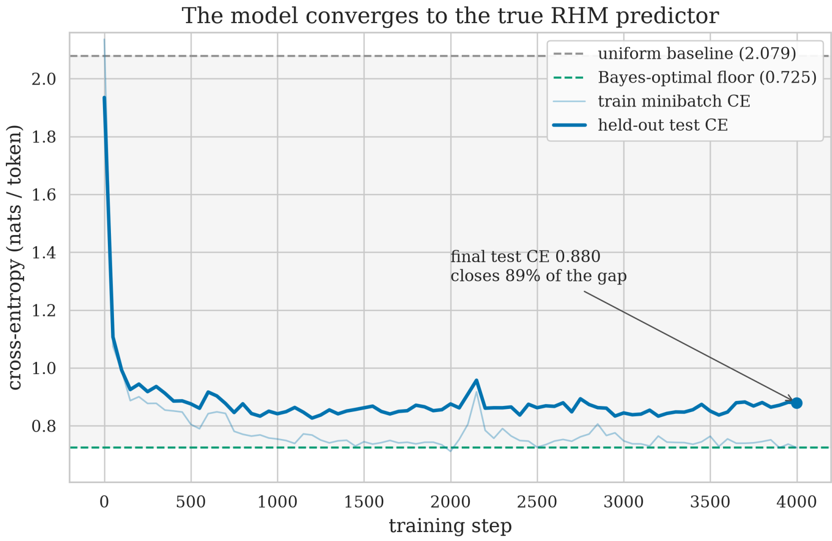

Figure 1: The transformer converges to the true RHM predictor. Held-out next-token cross-entropy (heavy line) falls from the uniform baseline ($\ln 8 = 2.079$, top dashed) toward the Bayes-optimal floor (0.725, bottom dashed) over 4000 training steps, ending at 0.880 nats — closing ≈89% of the uniform→Bayes gap (shaded). The faint line is the train-minibatch CE. Seeded reproduction of the pinned-arch canonical run (grammar seed 0, $n_{\text{layer}\{=\}}2, n_{\text{embd}\{=\}}128, n_{\text{head}\{=\}}4$), `results/refrun/`.

2.3. Pre-registered prediction

Following the project's honor-code rule, we committed our predictions to git **before** training any model or computing any probe (`PREREGISTRATION.md`). In brief, we predicted: (1) the root posterior is **linearly decodable** from the residual stream, with $R^2 > 0.5$ at the final layer and a shuffled baseline near 0; (2) the belief **blooms** — early positions cluster near the prior, late positions spread toward simplex vertices, with posterior entropy falling and activation radius growing with context k; (3) decodability **accumulates additively across depth**, lowest at the embedding and highest after the last layer; and (4) the representation is **causally used**, so steering the residual along a belief direction shifts the output toward the corresponding latent's leaves. We also registered alternative outcomes (degenerate collapse, non-linear-only encoding, MAP-only encoding, flat-with-depth) as falsification handles. We report against these predictions in Section 3.6.

3. Results

3.1. A single linear probe partially decodes the posterior

We fit one global least-squares affine map from the 128-d residual stream to the 8-class exact root posterior, on a probe set of $N = 400$ held-out examples (first, context-position agnostic, then wrt. to it), and score it by R^2 on held-out data. The final-layer probe reaches $R^2 = 0.38$, while at the control task (the same probe fit to **shuffled** labels) scores ≈ 0 (-0.085). So the posterior is linearly accessible above chance, but the probe explains only a third of the variance: the recovery is **partial**. Projected into a belief-space PCA basis (Figure 2), the probe's predicted posteriors occupy the same structured region as the exact posteriors and reproduce the same position gradient, but as a diffuse cloud rather than the discrete point set of the ground truth.[1] The affine correspondence is particularly weak for the root; it strengthens markedly for deeper latents (next subsection) and at later layers.

[1]The ground-truth panel looks sharper partly because the exact posterior takes few distinct values, so identical points overplot, whereas every probe prediction differs slightly.

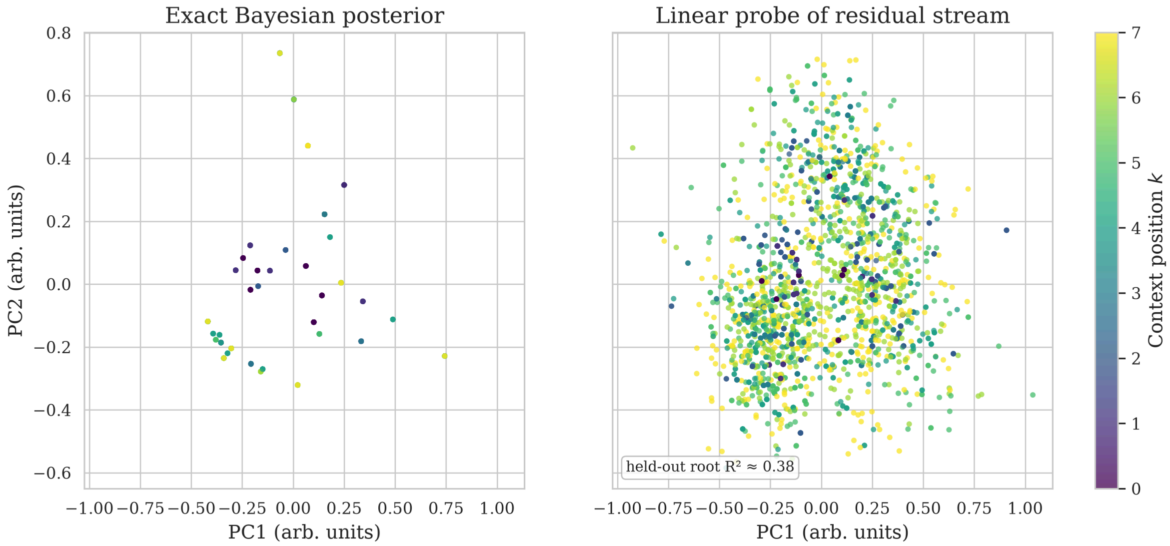

Figure 2: The residual stream is a **partial** affine image of the exact belief simplex. **Left:** PCA(2) of the exact root posteriors (few distinct values, hence sharp overplotted dots); small-k points sit near the prior, large-k points spread toward vertices. **Right:** the linear probe's predictions in the same basis, colored by context position — the same region and position gradient, but a diffuse cloud: held-out root $R^2 \approx 0.38$, an imperfect recovery, not an exact one.

3.2. Decodability accumulates across depth and context

The residual stream is a running sum of layer contributions, so we ask where the belief is built. Both panels here decode the **root** posterior. The residual stream is read at three points — the model has two attention blocks, so `output_hidden_states` returns $n_{\text{layer}} + 1 = 3$ snapshots: index 0 is the token+position embedding, index 1 is after block 1, index 2 is after block 2 (the "Layer 0/1/2" axis is these three readout points, not three transformer layers). Root-posterior probe R^2 rises along them: -0.00 at the embedding, 0.15 after block 1, 0.38 after block 2, while the shuffled control stays at ≈ 0 throughout (Figure 3, left) — each block adds belief-relevant signal. Resolving the same root probe by context position (Figure 3, right), the root is already decodable at early positions even at the shallow readout points, and at the final readout it is near-perfectly decodable ($R^2 = 1.0$) for the first few positions 0–3.[2].

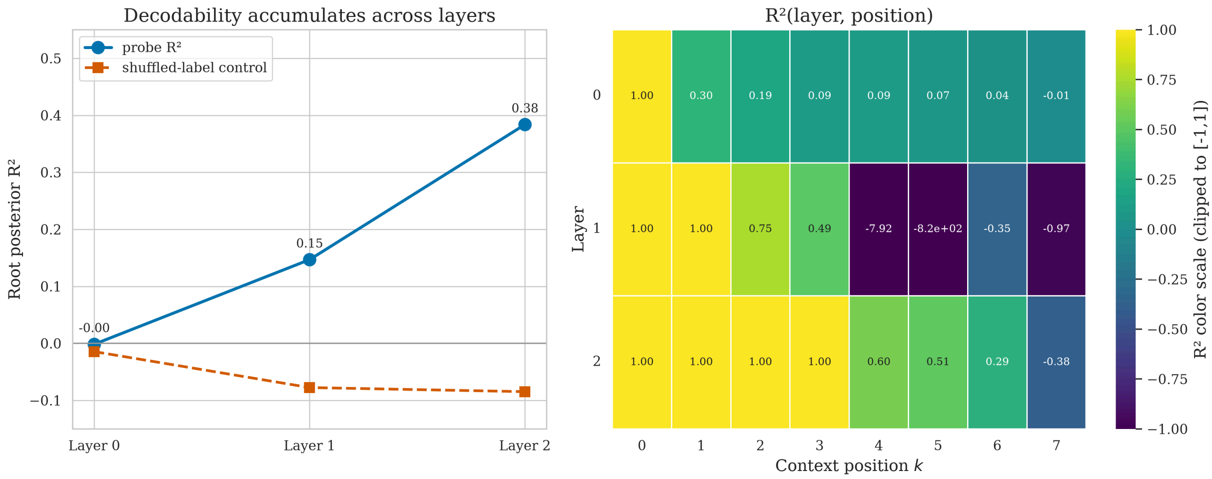

Figure 3: Root-posterior decodability accumulates across residual depth and context. Both panels decode the **root** class; "Layer 0/1/2" are the three residual readout points (embedding, after block 1, after block 2 = $n_{\text{layer}\{+\}}1$). **Left:** root-posterior R^2 rises along them ($-0.00 \to 0.15 \to 0.38$) while a shuffled control stays at ≈0. **Right:** R^2(readout point, position) heatmap; the final readout reaches $R^2 = 1.0$ on the earliest positions.

[2]A few mid-readout cells are strongly negative (the probe underperforms the mean predictor on those positions); the heatmap colors are clipped for legibility, but the cell labels show the raw values

3.3. The whole latent hierarchy is encoded — local latents most strongly

The root is only one of the tree's hidden latents. Probing the exact posterior over latents at **every** level reveals a clear gradient with some node-level variation (Figure 4). Precisely, root (L_0) $R^2 = 0.38$; the two level-1 mid latents 0.51 and 0.59; and the four deepest level-2 latents 0.61, 0.66, 0.50, and 0.66. The residual encodes the entire hierarchy, but the **local, near-leaf** latents, which are more directly predictive of the next token, are usually read off more cleanly than the coarse global root. Root is the **hardest** latent to decode, likely because it is the most abstract thus least locally predictive.

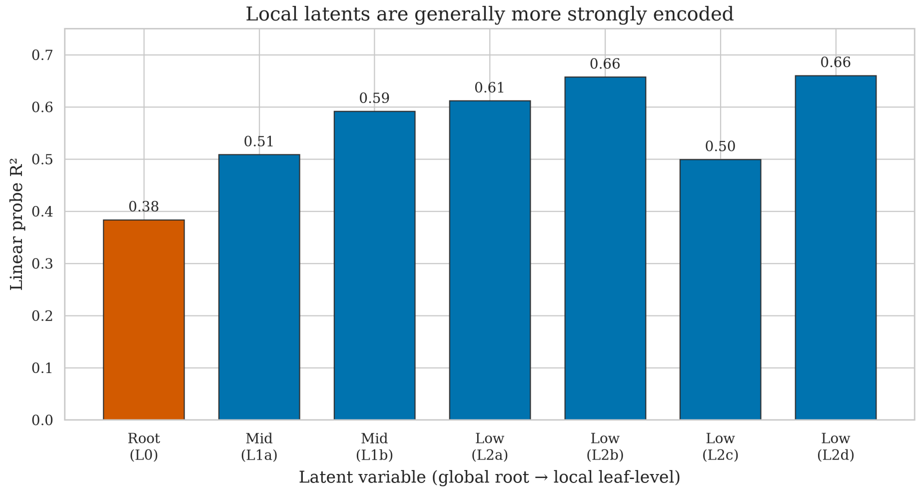

Figure 4: The residual encodes the whole latent hierarchy, deeper/local latents most strongly. Probe R^2 is lowest for the root ($L_0 = 0.38$, the spec's primary target), higher for level-1 $(0.51, 0.59)$, and generally higher for level-2 $(0.61, 0.66, 0.50, 0.66)$.

An important refinement based on robustness studies (Appendix 7). This "deeper decodes stronger" reading is **refined** once the tree is one level taller. Scaling the identical per-level probe to $L = 4$ (the depth study in Figure 17) shows the gradient is really an **inverted U**: the mid-level latents decode strongest, while **both** the coarse root and the most-local leaf-parents fall off. The headline is not "local beats global" but "mid beats both ends" — previewed here, quantified across 60 runs in the appendix.

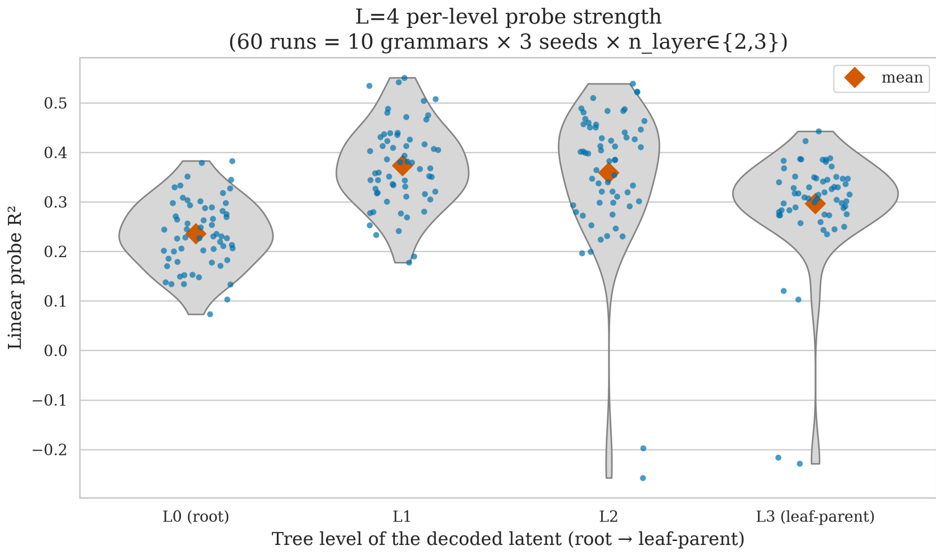

Figure 5: Preview of the inverted-U refinement (full detail in Figure 17). At $L = 4$, per-level probe R^2 across 60 runs rises from the root (L_0) to the mid latents (L_1, L_2) and falls back toward the leaf-parents (L_3) — an inverted U, not a monotone "deeper is stronger" ladder.

3.4. The belief blooms with context

Our central geometric result (Figure 6): projecting the linear belief readout into a belief-space PCA(2) basis, the points form a tight central cluster near the uniform prior when little context is observed and expand outward toward the simplex vertices as context accumulates. The mean radius about the prior anchor grows from 0.17 at $k = 0$ to ≈0.39 at $k = 7$, tracking the true posterior's own growth (0.17 → 0.47); equivalently, the exact posterior entropy falls with position (1.84 nats at $k = 0$ to 0.00 at $k = 7$). The right panel colors the same cloud by ground-truth root class: it is only **loosely** organized by class — consistent with the modest root decodability ($R^2 = 0.38$, MAP root accuracy 0.47, Figure 21) — not a clean separation. Raw residual PCA, dominated by token and position nuisance variance, does **not** bloom (radius–vs–k correlation ≈ 0.03, against ≈ 0.81 for the readout) — it is the **belief content** of the residual that does.

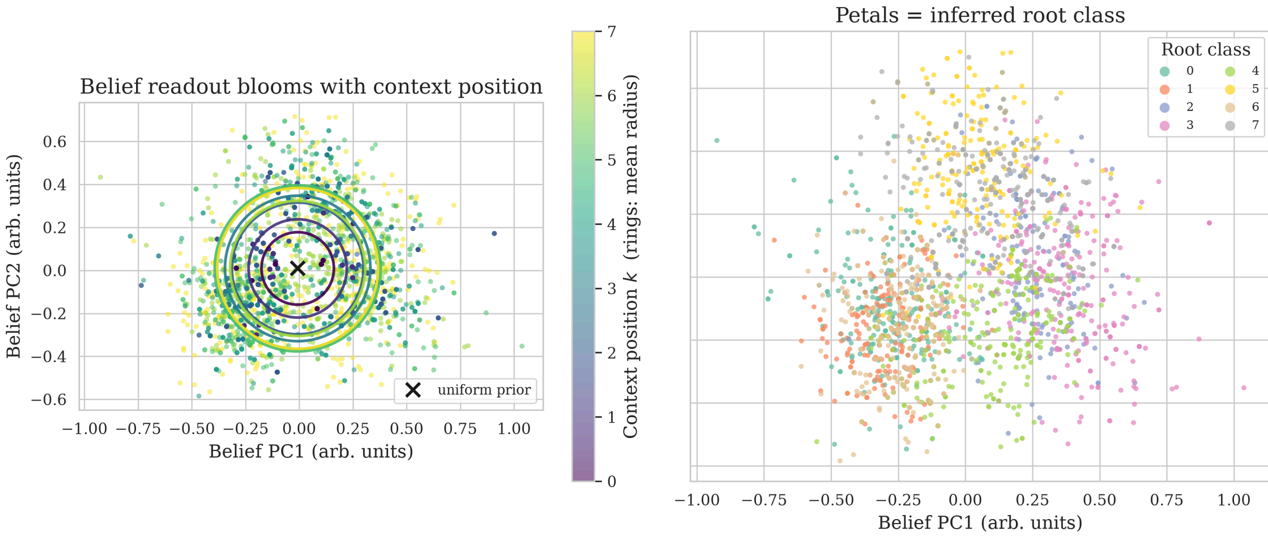

Figure 6: Headline result: the belief read out of the final-layer residual blooms with context. **Left:** points colored by context position k with rings marking each position's mean radius about the prior (×); the cloud expands outward as context grows (mean radius 0.17 → 0.39, tracking the true posterior 0.17 → 0.47). **Right:** the same cloud colored by ground-truth root class is loosely organized by class (partial, not clean separation — root MAP accuracy 0.47).

3.5. Causal steering: the belief is used, not just decodable

A representation can be decodable yet causally inert. To distinguish the two we apply a mean-difference patch to the layer-1 residual, pushing it by α times the belief direction toward a chosen latent, and measure the effect on the next-token distribution (Figure 7). Steering **toward the true** latent leaves predictions untouched (the model already holds that belief): the true token's log-probability stays at −0.73 for all α. Steering **toward a wrong** latent collapses the mean log-probability of the true next token from −0.73 to −8.5 as steering effort α grows, and re-routes probability mass onto the wrong latent's children — the next-token probability the model assigns to exactly the leaf symbols that the steered-to (wrong) latent can emit at that position (summed softmax mass over those symbols, averaged across positions and examples) rises from 0.22 to 0.53. The belief state is therefore causally used, not merely decodable.

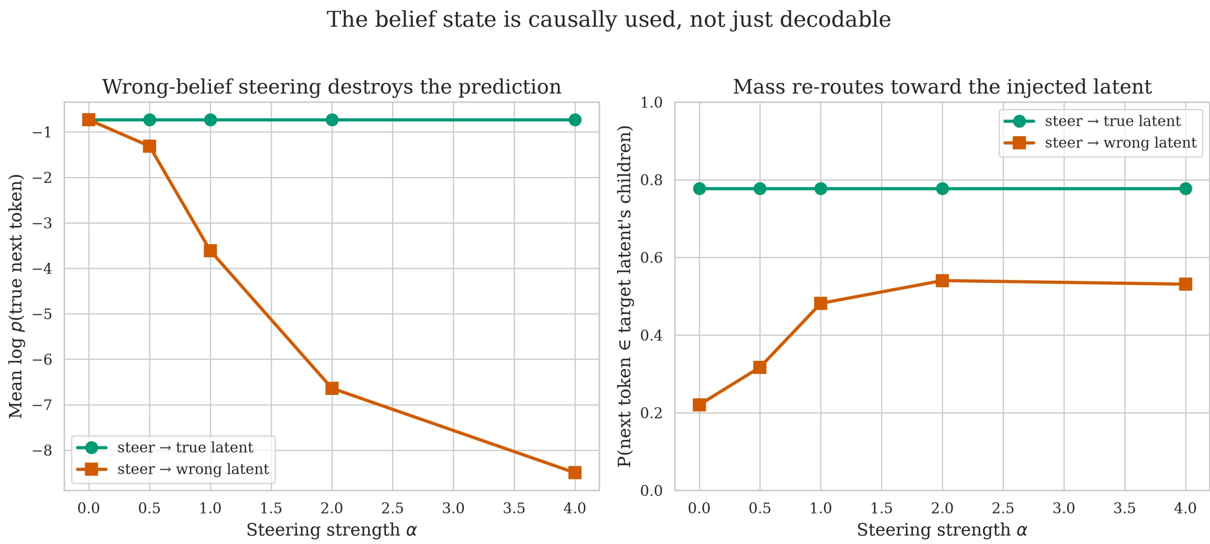

Figure 7: The belief state is causally used. Steering the layer-1 residual toward the **true** latent leaves predictions intact (green); steering toward a **wrong** latent collapses the true next token's log-probability (left, orange: $-0.73 \rightarrow -8.5$) and re-routes mass onto the wrong latent's children (right, orange: $0.22 \rightarrow 0.53$).

3.6. Pre-registration scorecard

We grade each pre-registered prediction honestly against the outcome (Table 2).

I was over-using monotonicity in my initial guess, which is a very strong claim, to be revised (shuoldn't have written it in the first place)

Prediction	Outcome	Verdict
Linear decodability, root $R^2 > 0.5$, shuffled ≈0	Root $R^2 = 0.38$ (shuffled ≈ 0). Decodable and well above baseline, but below the 0.5 threshold for the root; **mid/deep latents do exceed 0.5** (up to 0.66).	Partial
Blooming with context (radius ↑, entropy ↓)	Mean radius $0.17 \rightarrow 0.39$; posterior entropy $1.84 \rightarrow 0.00$.	Hit
Monotone layer-wise accumulation of R^2	$-0.00 \rightarrow 0.15 \rightarrow 0.38$ across the three residual indices.	Hit
Causal usage via steering (ordered effect)	Steering→wrong: log p collapses $-0.73 \rightarrow -8.5$, monotone in α; target-child mass $0.22 \rightarrow 0.53$.	Hit

Table 2: Scored pre-registered predictions (`PREREGISTRATION.md`, committed before any result). Three of four held; the lone miss is the root-specific $R^2 > 0.5$ magnitude.

Three of four predictions held cleanly. The single miss is the root R^2, which came in at 0.38 rather than the predicted > 0.5. We take this as a genuine and informative negative: the prediction was correct in **form** (linear, above baseline, growing with depth and context) but our magnitude estimate for the **root specifically** was optimistic. The latent-level sweep — which we did not pre-specify — explains why: the root is the coarsest, least locally-predictive latent, and the threshold we predicted is in fact met by every deeper latent in the hierarchy. None of the pre-registered failure modes (degenerate collapse, non-linear-only encoding, MAP-only encoding, flat-with-depth) materialized; in particular MAP root-class accuracy is only 0.47 while full-simplex R^2 is positive and graded, ruling out a MAP-only representation.

4. Engineering non-collapsing belief geometry

Everything so far lives on a tree whose belief, given the full leaf string, collapses to **certainty**: with uniform unambiguous rules the eight leaves invert level-by-level to exactly one root, so the $k = 8$ posterior is a delta and the belief path runs from the prior straight to a simplex vertex. The "interesting" mixing HMMs studied in the belief-geometry literature (Shai et al., 2026) behave oppositely — their belief never collapses to a single certain state; it instead traces a self-similar, fractal (Sierpinski-like) attractor that fills a structured region of the simplex interior, approaching the vertices in the limit without ever settling on one — but the RHM has no recurrence, so "slow forgetting" has no analog. A possible analog of "the state can't be restored" is a **non-invertible observation channel**: a grammar where even the full leaf string leaves the root uncertain, so the reachable-belief set is a non-trivial attractor in the simplex rather than a path to a vertex. This pass asks what it takes to engineer that, and whether the linear probe still recovers it.

We add two per-level knobs to the grammar (`rhm.py`, both reducing exactly to the canonical RHM at their off setting, verified byte-identical). **Ambiguity** ρ shares a fraction of the $v \cdot m$ rule entries' child-tuples **across parents** (many-to-one leaf→root structure); $\rho \in \{0, 0.3, 0.6\}$. **Skew** draws each parent's rule-choice probabilities non-uniform from a Dirichlet with concentration α: `none` is the canonical uniform $1/m$; `mid` uses $\alpha = 1.0$ and `high` uses $\alpha = 0.2$ (smaller $\alpha \implies$ more peaked, near-deterministic rule choice). Both belief propagation and the brute-force reference were generalized to **weighted** sum-product using the same probabilities the sampler uses; the BP↔ brute-force agreement holds to 2.5×10^{-16} under both knobs simultaneously. We sweep the full 3×3 grid $\times$ 3 rule-table draws — **27 runs** at the pinned published arch (`run_noncollapse.py`, writing `results/noncollapse/`), with the (none, none) cell reproducing the pass-1 baseline.

Mean exact $k = 8$ root-posterior entropy is exactly 0 at unambiguous setup regardless of the `skew`. It rises monotonically with ambiguity ρ growth: $0.00 \to 0.71 \to 1.38$ nats at `skew=none` (Figure 8, left). The full entropy-vs-k trajectory (Figure 8, right) makes the phenomenon legible — at $\rho = 0$ the belief collapses cleanly to 0 by $k = 8$ (the bloom-to-vertex), while at $\rho = 0.3$ and $\rho = 0.6$ it **plateaus** at a positive floor: a genuine non-collapsing attractor. Higher skew **worsens** the plateau but never removes it.

somewhat expected

The linear probe, surprisingly, sharpens. At every cell of the grid the residual stream still linearly encodes the (now spread) posterior well above the shuffled-label baseline of ≈ -0.07 (Figure 9). That baseline is the same probe refit to randomly **permuted** labels and scored on held-out data, averaged over the 27 runs — a near-zero (slightly negative) "no real signal" reference. The probe did not **degrade** as ρ rose: R^2 **rises** with ambiguity, $0.36 \to 0.42 \to 0.53$ at `skew=none`, and the mid- and low-level probes rise even more steeply ($L_1 : 0.46 \to 0.71$; $L_2 : 0.35 \to 0.82$). A delta-collapsed posterior is a near-constant target once the context pins it, starved of variance; a non-collapsing posterior is a richer, higher-variance, more linearly-structured signal, so linear decodability **improves**. The attractor comparison (Figure 10) confirms it: at $\rho = 0.6$ the probe readout traces the same structured interior cloud as the exact posterior, versus the vertex-bound bloom at $\rho = 0$.

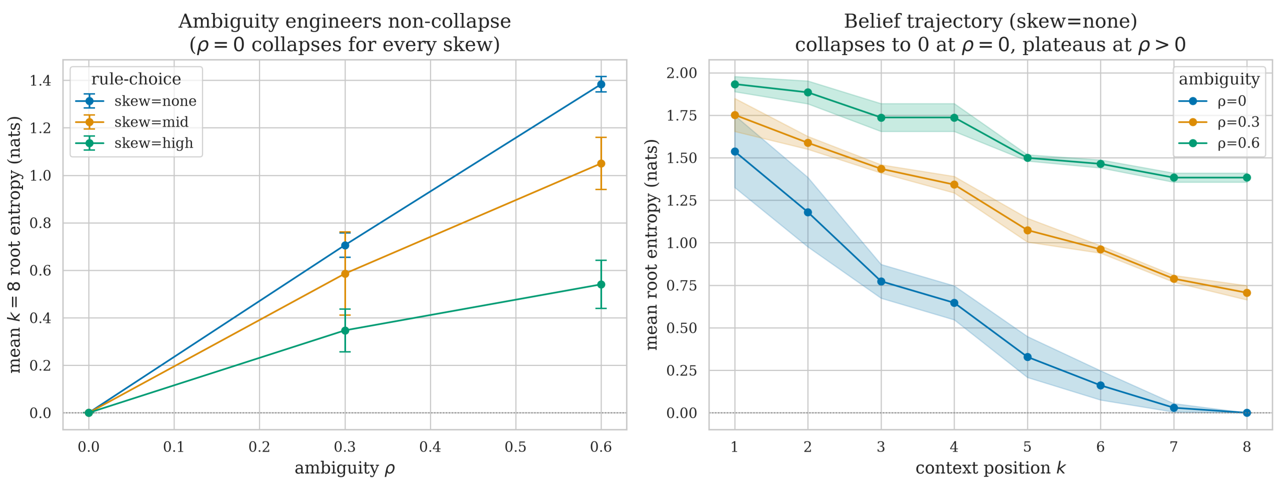

Figure 8: Ambiguity, not skew, engineers non-collapse. **Left:** mean exact $k = 8$ root posterior entropy vs ambiguity ρ, one line per skew (error bars over 3 draws). At $\rho = 0$ entropy is exactly 0 for **every** skew — skew alone does not prevent collapse — and it rises monotonically with ρ. **Right:** the full entropy-vs-context trajectory at `skew=none`; at $\rho = 0$ the belief collapses to 0 by $k = 8$, at $\rho > 0$ it plateaus at a positive floor — a non-collapsing attractor.

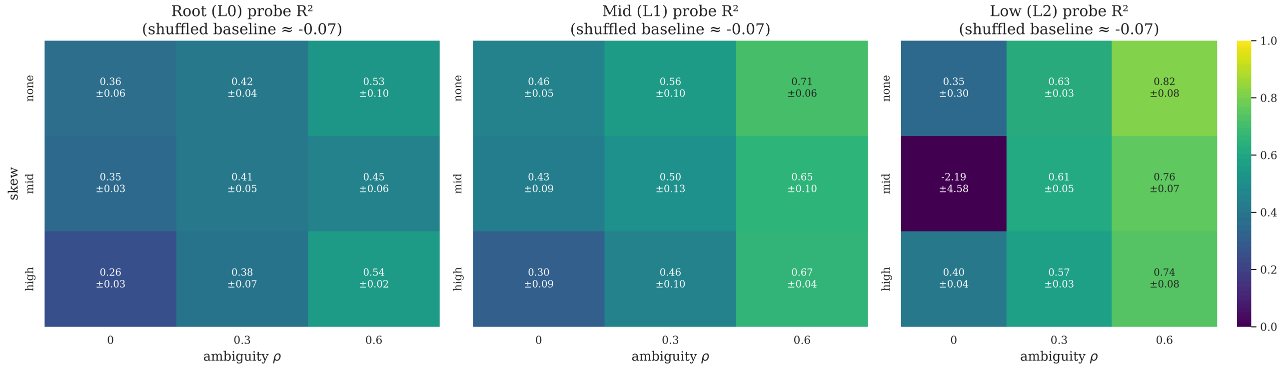

Figure 9: The linear probe survives ambiguity and skew, and **sharpens** with ambiguity. Per-level probe R^2 (root L_0, mid L_1, low L_2) over the 3×3 skew $\times$ ambiguity grid; each cell is mean $\pm$ SD over 3 rule-table draws, against a shuffled baseline of ≈ -0.07. R^2 rises left-to-right (with ρ) at every level. The single dark L_2 cell at $(\text{mid}, \rho\{=\}0)$ is a near-deterministic-target instability, not a decodability failure.

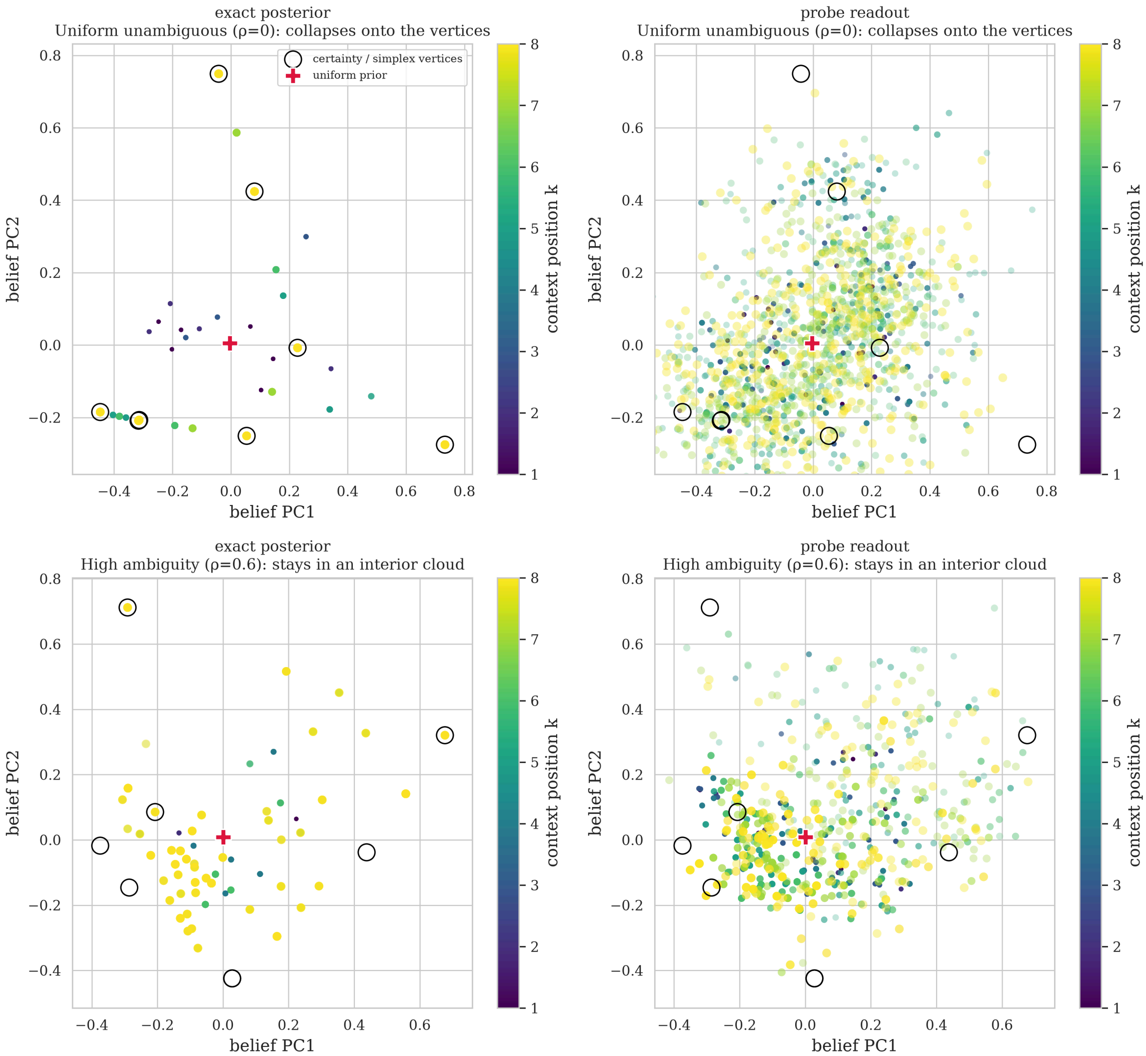

Figure 10: Exact posterior (left column) and linear-probe readout (right column) in belief-PCA, colored by context position k — marker size and opacity grow with k (the last token is largest and fully opaque, earlier context dimmed) so the endpoint of each trajectory stands out — for the uniform corner $\rho = 0$ (top) and the high-ambiguity corner $\rho = 0.6$ (bottom). Open black circles mark the eight certainty / simplex vertices (one-hot roots) and the red + the uniform prior, both projected into the panel's own basis (the circles are hollow so the points landing inside them stay visible). Because root-class labels are not aligned across grammars, each row keeps its **native** belief-PCA basis and axis limits are shared only **within** a row; the vertex markers give the label-agnostic reference. At $\rho = 0$ the late-context beliefs land **on** the vertices (collapse to certainty); at $\rho = 0.6$ they stay in an interior cloud near the prior, **off** the vertices (non-collapse) — and the probe readout occupies the same region as the exact posterior.

5. Discussion and limitations

The picture is coherent: a tiny transformer trained to near-Bayes-optimal loss on a hierarchical grammar carries a **partial** linear image of the posterior over the grammar's hidden latents in its residual stream — fuzzy for the coarse root ($R^2 = 0.38$), sharper for the local latents (R^2 up to 0.66). That image is assembled additively across layers, blooms from the prior toward the vertices as evidence accumulates, spans the full latent hierarchy, and is causally relied upon at generation time. We probe against an **exactly known** belief state (computed by belief propagation, not approximated), which is what lets us quantify the recovery as partial rather than merely assert it — but the linear recovery itself is imperfect, and we do not claim the probe reconstructs the posterior exactly. This reproduces the core Simplex belief-geometry phenomenology in a setting where the ground-truth belief state is known exactly.

Methodologically, this paper sits between the two reference points in the bibliography: Cagnetta et al.'s RHM study uses the same hierarchical data model but evaluates the last-token prediction setting, while

Shai et al.'s **Transformers learn factored representations** motivates pooling predictive vectors across contexts. Our readout follows the latter all-context-position spirit, treating every prefix position as a belief state to be decoded, while keeping Cagnetta et al.'s exact RHM grammar as the data source.

The most interesting wrinkle is the **inverted strength gradient**: the global root, the spec's nominal target, is the hardest latent to decode, while local near-leaf latents are read off most cleanly. This is intuitive in hindsight — next-token prediction rewards representing whatever is most locally predictive — but it sharpens the belief-geometry claim: the residual stream tracks the **full** latent posterior, weighted toward what the task needs, not a single privileged variable.

Limitations. (i) The grammar is deliberately tiny and fully enumerable ($s = 2$, $L = 3$, 1024 trees); this is what makes the beliefs exact and the verification airtight, but it leaves open how the geometry scales to larger, non-enumerable RHMs. (ii) At $L = 3$ the root posterior collapses to certainty by $k = 3$ on many strings, compressing the dynamic range over which blooming is visible for the root; the spec anticipated this and suggested $L = 4$ as a follow-up. (iii) The strongly-negative mid-layer probe cells indicate the affine probe is locally mis-specified at some position/layer combinations; a per-position or whitened probe would tighten these estimates. (iv) Steering is applied at a single layer (layer 1) along a mean-difference axis; a learned causal direction and a layer sweep would strengthen the causal claim. (v) The detailed pass-1 figures (Figure 2–Figure 7) are from a single training run and grammar draw; the grammar-sweep and architecture-sweep sections quantify how the **headline** metrics move across 10 grammars / 3 seeds and across 11 capacity configs / 3 grammars / 3 seeds respectively, but the architecture sweep is one-axis-at-a-time (no $n_{\text{layer}} \times n_{\text{embd}} \times n_{\text{head}}$ interaction cells). The $L = 4$ section extends the family one tree level deeper; still-larger grammars ($L \geq 5$, larger v, $m > 2$), where brute-force belief verification becomes intractable, remain a future pass.

6. Conclusion

On an exactly-solvable hierarchical grammar, a small transformer's residual stream is a linear image of the exact Bayesian belief simplex: it accumulates across depth, blooms with context, encodes the whole latent hierarchy (local latents most strongly), and is causally used.

7. Appendix: Robustness studies

summarize these studies and main findings in a paragraph.

7.1. Generalization across grammars

The results above come from a single grammar[3] and a single training run. To test whether they are properties of the RHM **family** under the same fixed constraints[4] rather than artifacts of one rule draw, we sweep the content of the rule table[5] across ten independently sampled grammars with three replicate training seeds each, resulting in 30 runs. We have ensured that the sampled grammars aren't isomorpic by post-factum ensuring that the respective nodes adjacency tables are not equivalent up to a per-level symbol permutation.[6] The model architecture / training code remained the same as in previous experiments.[7].

- **All grammars train to near-Bayes.** Every run closes 0.82– 0.93% of the uniform to Bayes-optimal loss gap (mean 0.89 ± 0.03), in line with the initial observation. The runs details can be seen in Table 3.
- **The root remains the weakest-decoded level** (Figure 11). Pooling probe R^2 by tree level, the root (L_0) averages 0.35 ± 0.07 (range 0.24–0.48), well below the mid (L_1, mean 0.50) and leaf-parent (L_2, mean 0.49) latents. The deepest level shows the widest spread (one grammar dips slightly negative).

investigate; not a failed training. poor train-test split?

node-level rule structure matters most for the most local latents?

[3] `grammar = 0`

[4] Reminder: fixed constraints so far are ($s = 2$, $L = 3$, $v = 8$, $m = 2$, uniform unambiguous rules)

[5] See the example (first tried grammar) rule table at Table 1

[6] Test code in `grammar_iso.py`

[7] Reminder: model architecture so far: GPT-2-style transformer, 2 layers of 4 attn heads each, hidden size 128, 4k train steps. A single master seed controls model initialisation, training-data sampling, probe sampling, and the probe split, so replicates measure end-to-end sampling variance. Each run writes a deterministic, self-contained directory (`results/sweep/g{NN}_L2_d128_h4_s{S}/`) holding its grammar, a `config.json` manifest (resolved config, git SHA, timestamp), and per-run metrics — recoverable for this paper independent of any experiment-tracker.

- **Blooming and belief-sharpening alongside context hold for every grammar** (Figure 12). The exact posterior entropy falls with context position k in all ten grammars, and the belief readout's radius grows with context.
- **Causal steering replicates with a stable effect size** (Figure 13). Steering the layer-1 residual toward a wrong latent collapses the true next token's mean log-probability in every run: as the patch strength α increases from 0 to 4, that log-probability drops by 6.9 ± 0.5 nats (mean $\pm$ SD across the 30 runs), with low run-to-run variance.

In short, all four earlier findings – poor root decodability, blooming, layer-wise accumulation (recomputed in every run's per-layer probe), and causal steering – replicate across the grammar family.

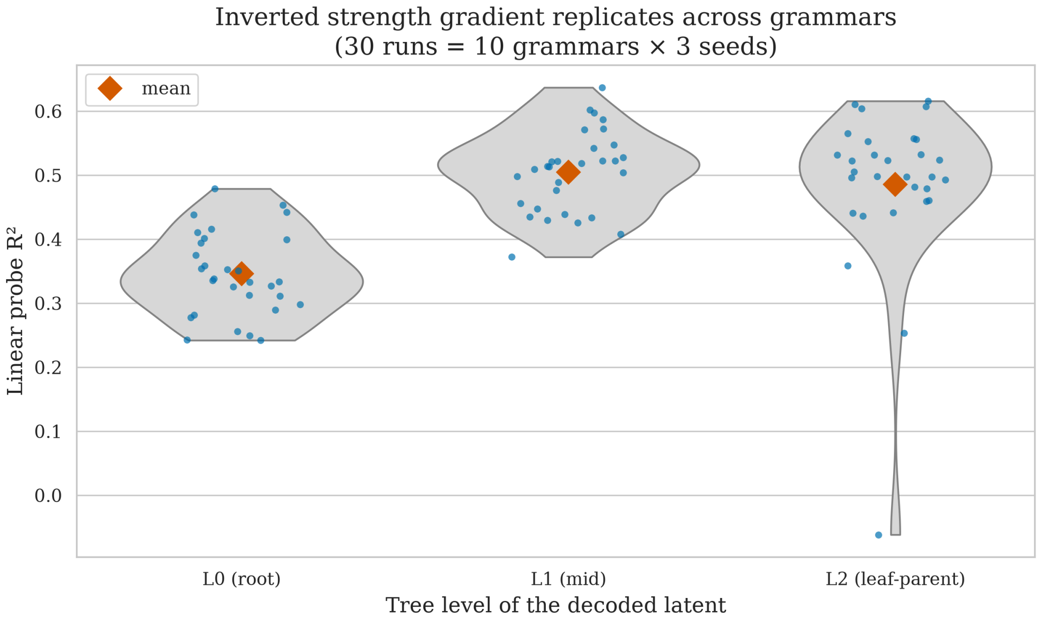

Figure 11: The inverted strength gradient replicates across the grammar family. Each point is one of 30 runs (10 grammars × 3 seeds); violins show the per-level distribution of probe R^2, diamonds the means. The root (L_0) is the weakest-decoded level (0.35 ± 0.07), below the mid (L_1) and leaf-parent (L_2) latents (≈ 0.49–0.50).

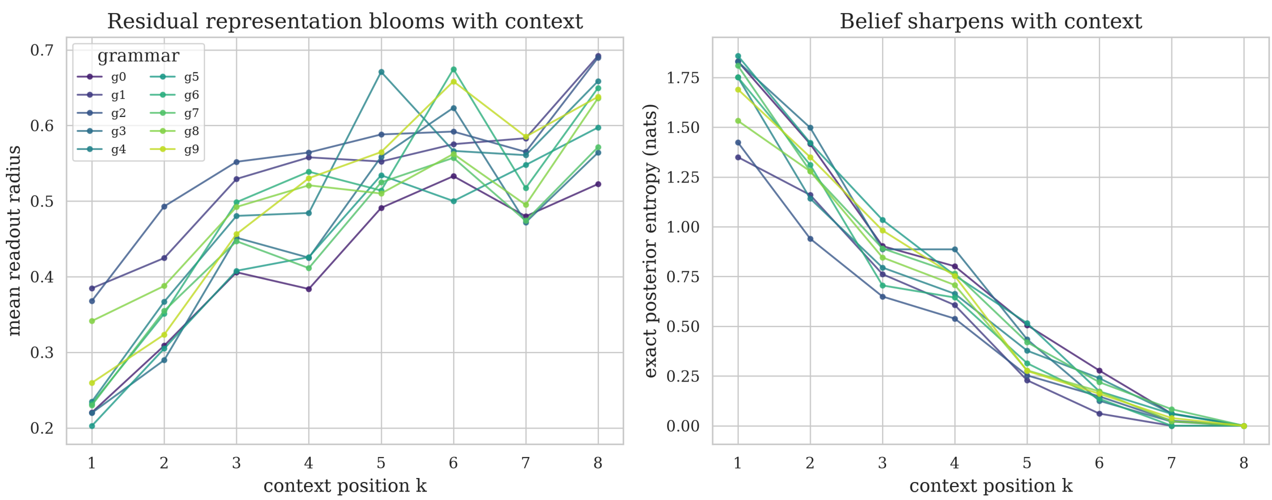

Figure 12: Blooming is family-wide. **Left:** the belief readout's mean radius grows with context position k (one line per grammar, averaged over seeds). **Right:** the exact posterior entropy falls with k for every grammar.

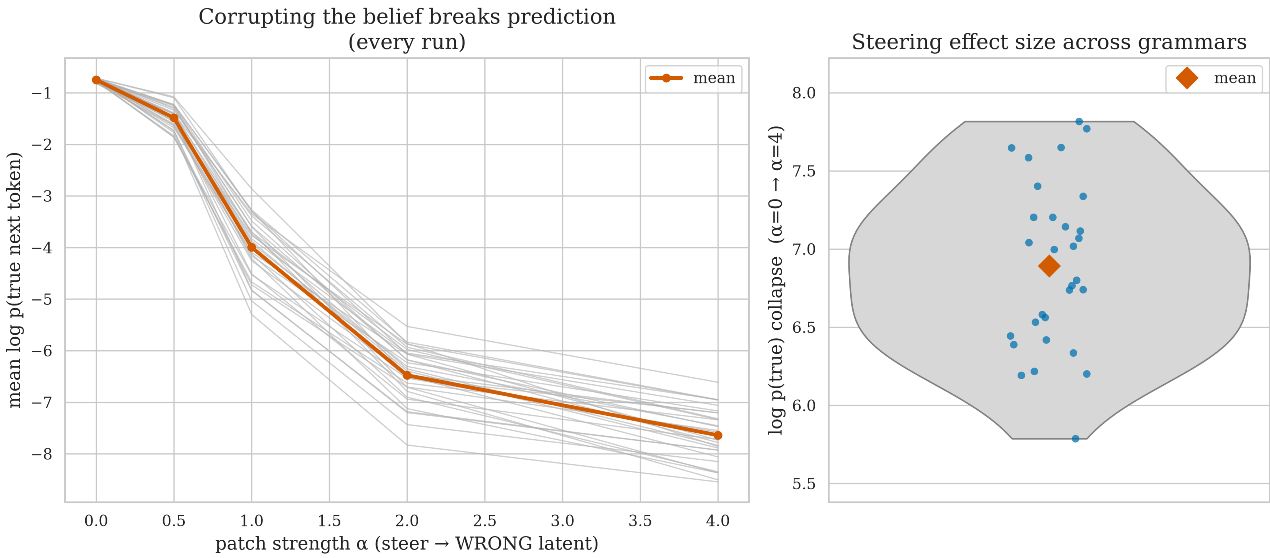

Figure 13: Causal steering replicates with a stable effect size. **Left:** steering the layer-1 residual toward a **wrong** latent collapses the true next token's log-probability in every run (grey lines), mean in orange. **Right:** the distribution of the collapse magnitude (grey violin, orange diamond = mean) — the drop in the true token's mean log-probability as $\alpha\{=\}0 \rightarrow \alpha\{=\}4$ — across all 30 runs: 6.9 ± 0.5 nats.

7.2. Architecture dependence

The grammar sweep fixes the architecture and varies the data; this section does the opposite. Again, we keep the RHM constraints. We vary the transformer's capacity one axis at a time around the baseline. We consider (in bold are baseline params) n_{layer} in {1, **2**, 3, 4}, n_{embd} in {16, 64, **128**, 256}, n_{head} in {1, 2, **4**, 8}, and training steps in {**4000**, 16000}.

Each axis varies independently with the other three pinned at baseline, the baseline being their common center. Every config is run across `grammar` $\in \{0, 1, 2\}$ and `seed` $\in \{0, 1, 2\}$ from grammar sweep, **99 runs total**[8]. Small models are **expected** to underfit.

Model capacity lifts decodability, with a low-width floor (Figure 14). Width is the strongest lever, across $n_{\text{embd}} = 16 \rightarrow 256$ the root probe R^2 climbs monotonically (from being undecodable at 16): $0.07 \rightarrow 0.21 \rightarrow 0.35 \rightarrow 0.44$, with the mid ($0.12 \rightarrow 0.57$) and leaf-parent ($0.15 \rightarrow 0.59$) levels rising in parallel. Depth lifts every level too: across $n_{\text{layer}} = 1 \rightarrow 4$ the root goes $0.22 \rightarrow 0.42$, the mid $0.32 \rightarrow 0.54$, and the leaf-parent $0.34 \rightarrow 0.48$. Number of heads is the weakest axis: from 1 to 8 heads the root moves only $0.25 \rightarrow 0.36$, the mid $0.37 \rightarrow 0.48$, and the leaf-parent level is essentially flat ($0.42 \rightarrow 0.44$).

Root remains the weakest latent. At all eleven configs the root (L_0) is the weakest-decoded level.

Model depth stretches the build-up and lifts the readout (Figure 15). Plotting root R^2 against normalized residual depth (residual index over n_{layer}), every model rises from ≈ 0.01 at the embedding to its final-layer readout, and deeper models both stretch the accumulation curve and reach a higher endpoint: final-layer root R^2 is $0.22 \rightarrow 0.35 \rightarrow 0.39 \rightarrow 0.42$ for $n_{\text{layer}} = 1$ to 4, respectively.

Longer training has less effect on leaves decodability than on other latents. Quadrupling the training budget ($4000 \rightarrow 16000$ steps) lifts the root by +0.05 ($0.35 \rightarrow 0.41$) and the mid level by +0.05 ($0.48 \rightarrow 0.53$), but the leaf-parent level by only +0.02 ($0.44 \rightarrow 0.47$).

Decodability decouples from loss fit (Figure 16). Across all 99 runs the correlation between `loss_gap_closed` and probe R^2 is modest — 0.46 for the root, 0.42 for the mid, 0.17 for the leaf-parent, and the scatter is wide. A model can reach near-Bayes loss without linearly representing the coarse belief, and vice versa .

interesting

[8] `run_arch.py` , writing steps-disambiguated dirs `results/arch/g{NN}_L{n}_d{d}_h{h}_t{steps}_s{S}/`)

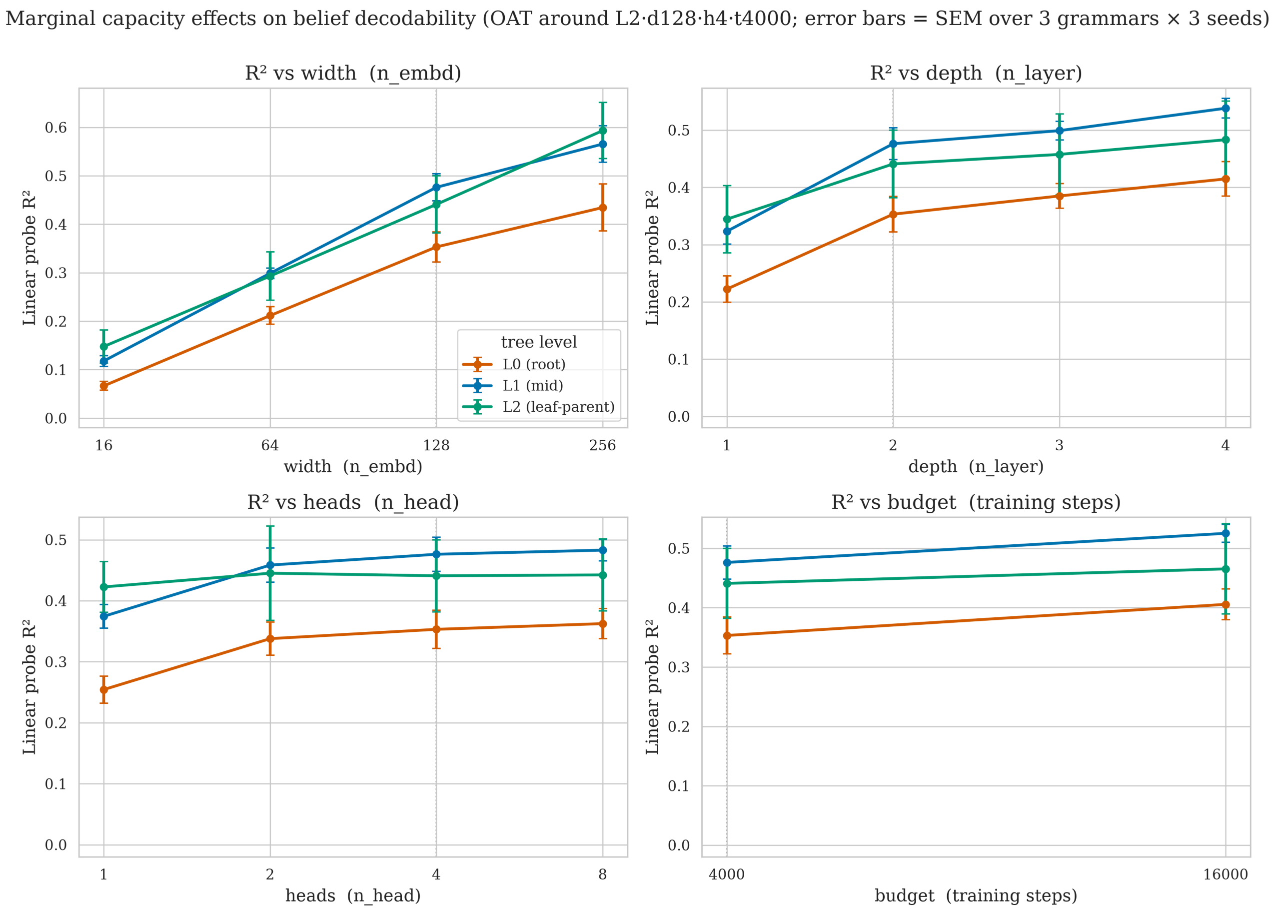

Figure 14: Marginal capacity effects. Each panel varies one axis with the other three at baseline; points are root (L_0), mid (L_1), and leaf-parent (L_2) probe R^2, error bars are SEM over 3 grammars × 3 seeds, the dotted line marks the shared baseline. Width and depth lift all levels (root collapses at $n_{\text{embd}} = 16$); heads matter least; the root is the lowest level in every panel.

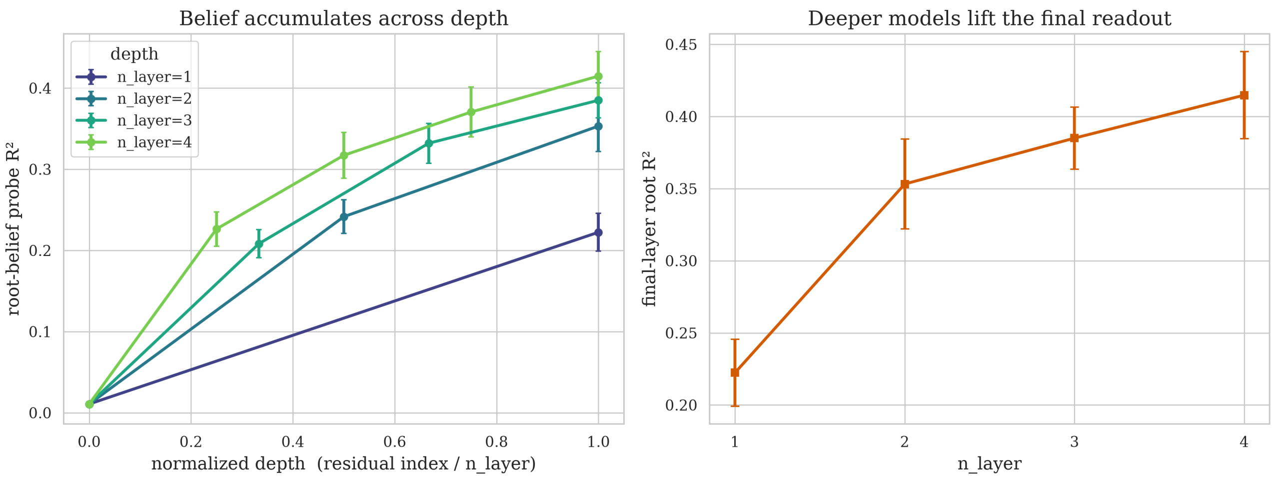

Figure 15: Depth and accumulation. **Left:** root-belief probe R^2 vs normalized residual depth, one line per n_{layer} (mean ± SEM over grammars × seeds); every model accumulates from the embedding to its readout, and deeper models reach higher. **Right:** final-layer root R^2 rises monotonically with depth.

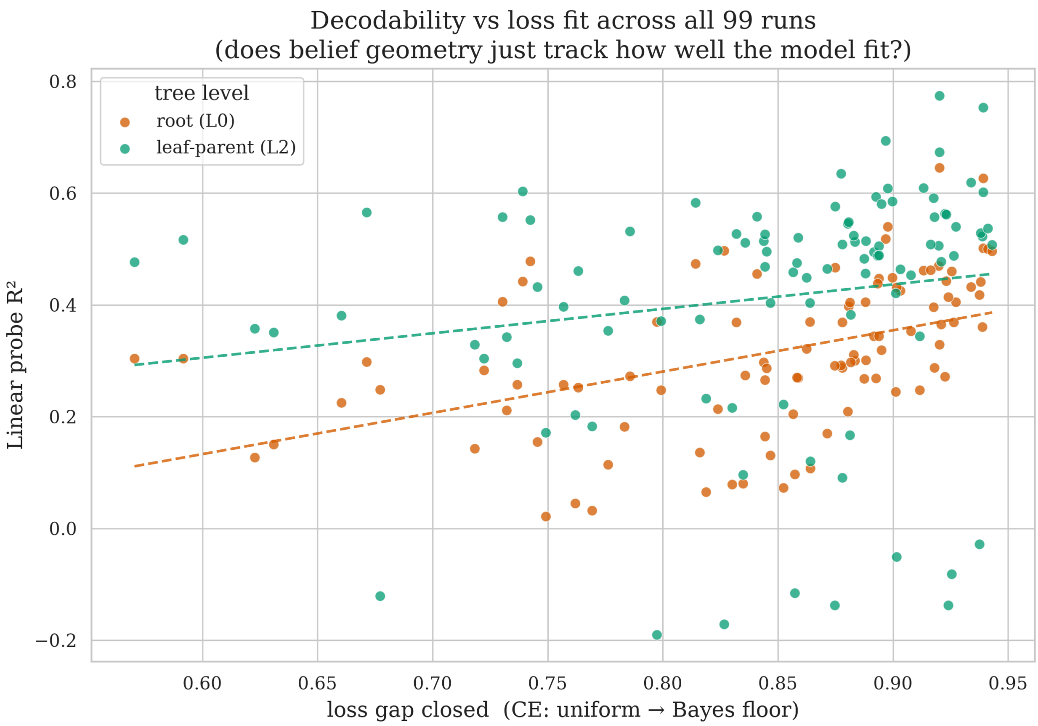

Figure 16: Decodability vs loss fit, all 99 runs. Root (L_0) and leaf-parent (L_2) probe R^2 against `loss_gap_closed` (fraction of the uniform→Bayes CE gap closed). Dashed lines are least-squares fits; the wide vertical spread at fixed loss fit shows decodability is not determined by how well the model fit the loss.

7.3. Going deeper: $L = 4$

The $L = 3$ tree is shallow enough that the root collapses to certainty within a few tokens, compressing the window over which its belief could be seen to bloom. Let's give the **root** more dynamic range. This pass increments the tree level, so context length becomes 16, and we deal with **fourth** latent level. Other HRM parameters remain the same.[9] We consider ten **non-isomorphic** grammars and three master seeds again. We vary model depth $n_{\text{layer}} \in \{2, 3\}$. Overall we make **60 runs**[10]. These run close 0.96–0.99 of the uniform→Bayes loss gap (mean 0.99) and are reported in full in Table 5.

linkage to ref

- **All earlier findings replicated at $L = 4$.** Linear decodability growth held (every level is read off well above the shuffled baseline of ≈ -0.04); the belief sharpens with context (exact root posterior entropy falls 1.86 → 0.00 monotonically over the 16 positions, Figure 18); belief **accumulates across layers** (root probe R^2 rises $-0.01 \to 0.07 \to 0.19$ for $n_{\text{layer}} = 2$ and $-0.01 \to 0.06 \to 0.17 \to 0.28$ for $n_{\text{layer}} = 3$, the deeper model will reach a higher readout, Figure 19); and causal steering still works (Figure 20) .

Root decodability becomes even weaker at $L = 4$ than at $L = 3$: the $n_{\text{layer}} = 2$ family mean falls to 0.19 (from 0.35 at $L = 3$), and even adding a third layer lifts it only to 0.28, still below the $L = 3$ value of 0.35.

unclear what is written here

The extra depth gives the **root specifically** room to bloom. At $L = 3$ the root collapsed to certainty within about three tokens, so its own belief barely had a window to expand (the $L = 3$ blooming we showed earlier is dominated by the local latents); the longer 16-position $L = 4$ context stretches that window out. Over it the exact **root** posterior entropy decreases smoothly and monotonically from 1.86 nats to 0, and the root belief readout radius grows overall from 0.18 to 0.50 (Figure 18). The radius growth is noisier than the local latents' — it wobbles in the back half — but the root now visibly sharpens with context rather than snapping to certainty.

The per-level gradient takes the inverted U-shape (Figure 17). Family-mean probe R^2 is $L_0 = 0.24$, $L_1 = 0.37$, $L_2 = 0.36$, $L_3 = 0.30$: it **rises** from the root to the mid-levels and then **falls** back toward the

[9] ($s = 2, v = 8, m = 2$, uniform sampling, unambiguous trees)

[10] `run_depth4.py`, writing `results/depth4/g{NN}_L{n}_d128_h4_t4000_s{S}/`, a separate root so the depth-agnostic dir names never collide with previous passes)

leaf-parents. Only 6 of 60 runs show the strict $L_0 < L_1 < L_2 < L_3$ ordering. Root remains the weakest decodable latent.

Figure 17: $L = 4$ per-level probe strength across the family. Each point is one of 60 runs (10 grammars × 3 seeds × $n_{\text{layer}} \in \{2, 3\}$); violins show the per-level R^2 distribution, diamonds the means. The gradient is an **inverted-U** — the mid-levels (L_1, L_2) decode strongest, the root (L_0) weakest, the leaf-parents (L_3) intermediate — not the pre-registered monotone ladder.

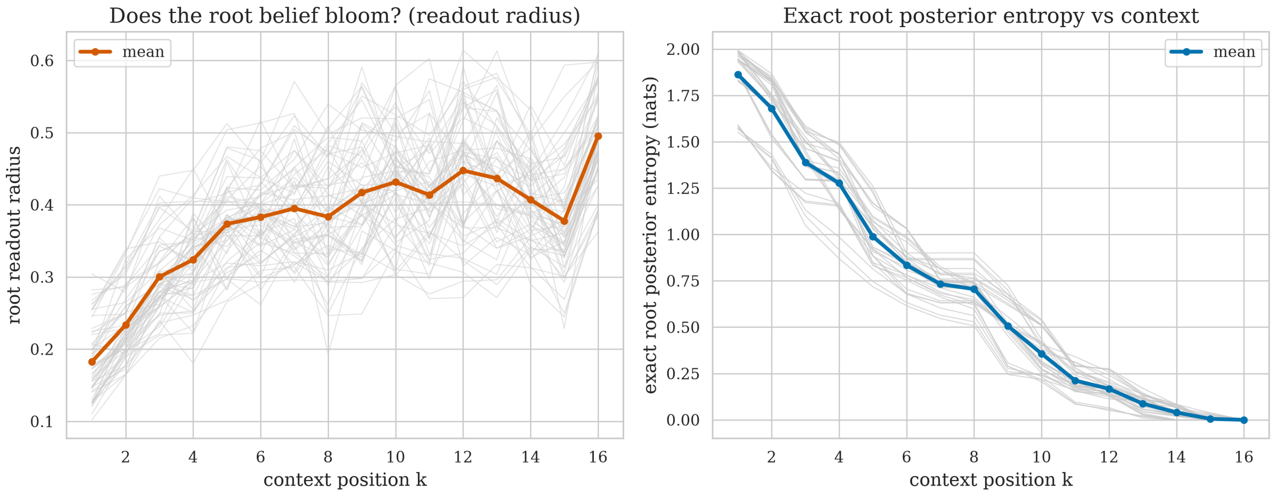

Figure 18: The root **does** bloom at $L = 4$. **Left:** the root belief readout radius grows with context position k over the 16-position window (grey: 60 runs; orange: mean), though non-monotonically in the back half. **Right:** the exact root posterior entropy decreases smoothly and monotonically from 1.86 nats to 0.

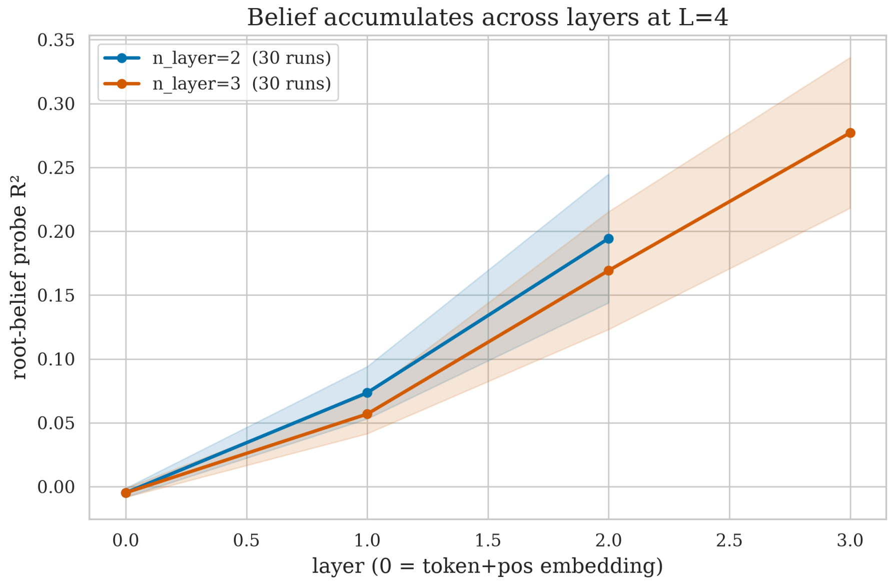

Figure 19: Belief accumulates across layers at $L = 4$. Root-belief probe R^2 vs residual index, one line per n_{layer} (mean $\pm$ SD over 30 runs each). Both archs rise monotonically from the embedding to the readout; the 3-layer model reaches a higher endpoint (0.28 vs 0.19).

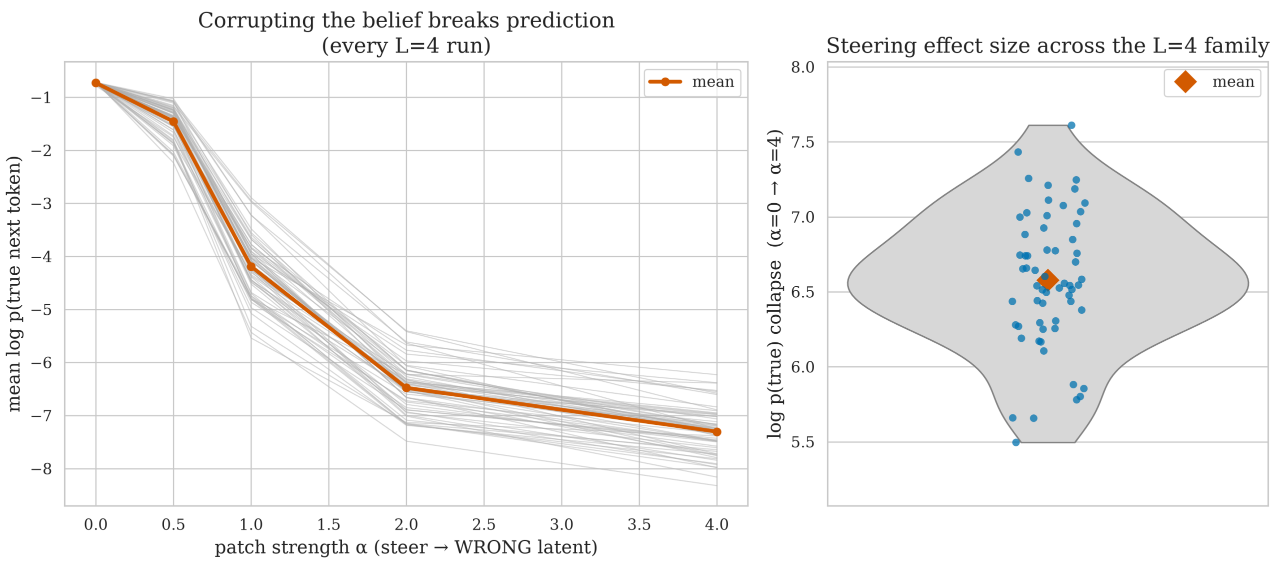

Figure 20: Causal steering replicates at $L = 4$. **Left:** steering the layer-1 residual toward a **wrong** latent collapses the true next token's log-probability in every run (grey), mean in orange. **Right:** the collapse magnitude (grey violin, orange diamond = mean) for $\alpha\{=\}0 \to \alpha\{=\}4$ across the 60 runs: 6.6 ± 0.5 nats.

8. Appendix: Root reconstruction diagnostics

The root-posterior probe is a regression target, but it is also useful to ask the harsher classification question: does the largest coordinate of the linear belief readout recover the sampled root class? Across 400 probe examples and 8 context positions (3200 example-position points), the answer is yes for 1565 points (48.91%). This is well above random guessing, but random guessing is only $1/8 = 12.5\%$ because there are eight root classes; the relevant baseline is not 50%.

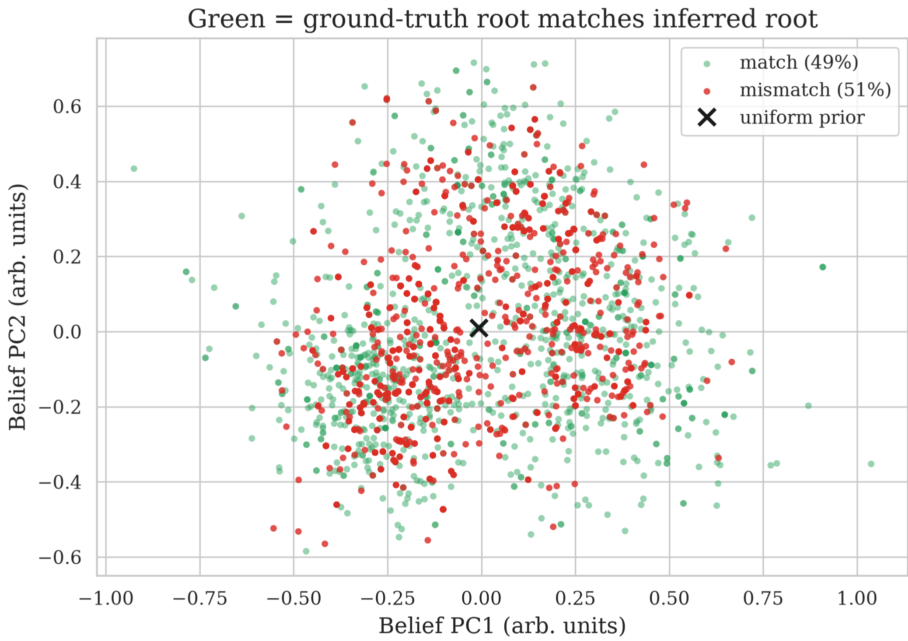

Figure 21: Root reconstruction from the linear belief readout. Green points are example-position pairs where the readout's argmax matches the ground-truth root; red points are mismatches. The overall match rate is 48.91%, compared with a random-guessing baseline of 12.5% (not 50%) for eight root classes.

Restricting to positions where nearly all evidence is visible makes the diagnostic sharper. With seven of eight symbols observed ($k = 6$), the exact Bayesian posterior's MAP root already matches the sampled root in 95.00% of examples, while the linear readout matches in 62.25%. With all eight symbols observed ($k = 7$), the exact posterior is deterministic, but the readout still reaches only 70.50%. Thus early ambiguity explains part, but not all, of the fuzzy root reconstruction.

Subset	**Prefix length**	**Readout match**	**Exact MAP match**
All example-position points	1-8	$\frac{1565}{3200} = 48.91\%$	–
All but last symbol	7	$\frac{249}{400} = 62.25\%$	$\frac{380}{400} = 95.00\%$
Full sequence	8	$\frac{282}{400} = 70.50\%$	$\frac{400}{400} = 100.00\%$

9. Appendix: The rule table as a graph

Table 1 lists the frozen grammar as text, three columns of `or`-separated child pairs. The same information is easier to scan as a graph: four columns of eight nodes (root, level 1, level 2, leaves), each parent's two rules drawn as two distinctly colored pairs of arrows into the level below. No edge skips a level, which is the "no cross-level ambiguity" property by construction.

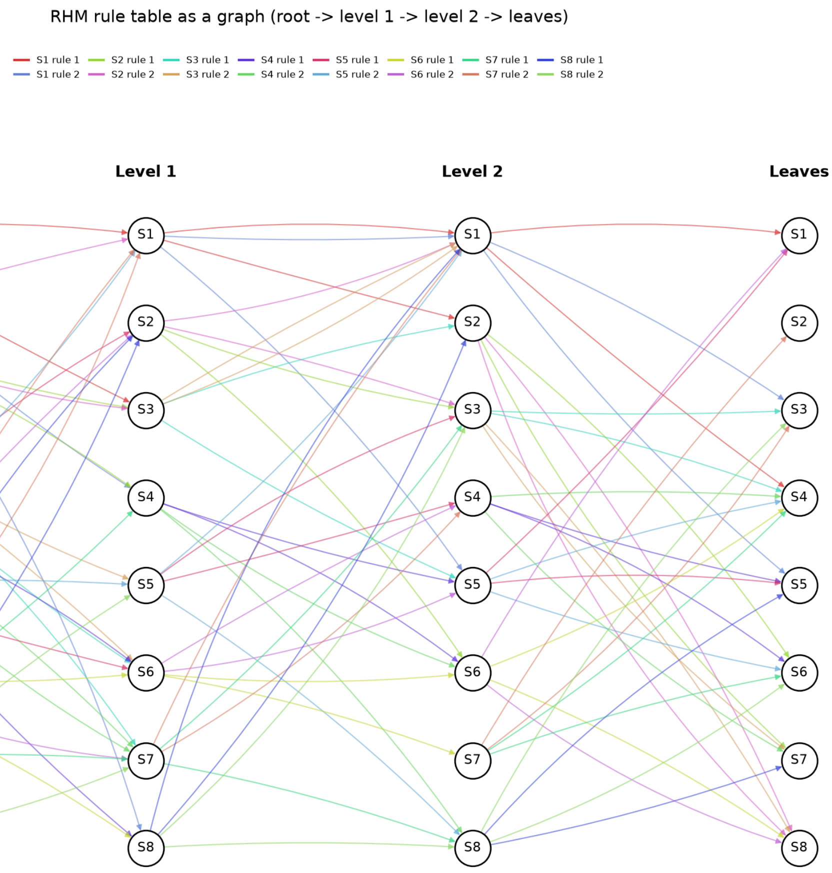

Figure 22: The frozen rule table of Table 1 redrawn as a DAG. Each of the eight symbols `S1`-`S8` appears once per level; its two production rules are drawn in two distinct colors (one per rule), chosen so that neither a node's own two colors nor neighboring nodes' colors are easily confused.

10. Appendix: Per-run grammar-sweep sanity table

Every run in the grammar sweep, no filtering. `grammar` indexes the rule-table draw (`Grammar.random(seed=grammar)`); `seed` is the master replicate seed; the architecture is pinned. `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` is the fraction of the uniform→Bayes gap closed (sanity covariate, not a gate); `root_r2` and `deepest_r2` are the level-L_0 and mean level-L_2 probe R^2. Loaded directly from `figures/sweep_sanity.csv`.

grammar	seed	n_layer	n_embd	n_head	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	2	128	4	0.931	0.848	0.242	0.498
0	1	2	128	4	0.891	0.878	0.243	0.497
0	2	2	128	4	0.918	0.858	0.281	0.532
1	0	2	128	4	0.847	0.906	0.375	0.556
1	1	2	128	4	0.812	0.932	0.479	0.616
1	2	2	128	4	0.845	0.908	0.353	0.496
2	0	2	128	4	0.905	0.886	0.402	-0.061
2	1	2	128	4	0.858	0.922	0.454	0.479
2	2	2	128	4	0.866	0.916	0.354	0.46
3	0	2	128	4	0.845	0.91	0.336	0.253
3	1	2	128	4	0.912	0.86	0.333	0.611
3	2	2	128	4	0.92	0.854	0.311	0.607
4	0	2	128	4	0.925	0.876	0.442	0.505
4	1	2	128	4	0.884	0.907	0.416	0.497
4	2	2	128	4	0.909	0.888	0.29	0.442

grammar	seed	n_layer	n_embd	n_head	test_ce	loss_gap_closed	root_r2	deepest_r2
5	0	2	128	4	0.853	0.911	0.25	0.461
5	1	2	128	4	0.841	0.92	0.277	0.482
5	2	2	128	4	0.848	0.914	0.334	0.493
6	0	2	128	4	0.868	0.896	0.351	0.523
6	1	2	128	4	0.882	0.885	0.4	0.553
6	2	2	128	4	0.836	0.919	0.312	0.523
7	0	2	128	4	0.964	0.82	0.298	0.441
7	1	2	128	4	0.909	0.861	0.256	0.359
7	2	2	128	4	0.892	0.874	0.326	0.436
8	0	2	128	4	0.897	0.863	0.338	0.524
8	1	2	128	4	0.838	0.906	0.359	0.532
8	2	2	128	4	0.868	0.884	0.327	0.532
9	0	2	128	4	0.899	0.894	0.411	0.557
9	1	2	128	4	0.988	0.827	0.394	0.565
9	2	2	128	4	0.87	0.916	0.438	0.604

Table 3: All 30 grammar-sweep runs (10 grammars × 3 seeds), pinned architecture, full 4000-step budget. No convergence filtering.

11. Appendix: Per-run architecture-sweep sanity table

Every run in the architecture sweep, no filtering — all 99 (11 one-axis-at-a-time capacity configs × 3 grammars × 3 seeds). `n_layer`, `n_embd`, `n_head`, `steps` give the capacity config; `grammar` and `seed` the replicate. `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` the fraction of the uniform→Bayes gap closed (covariate, not a gate); `root_r2` and `deepest_r2` the level-L_0 and mean level-L_2 probe R^2. The deliberately-underpowered small models (e.g. $n_{\text{embd}} = 16$) are included. Loaded directly from `figures/arch_sanity.csv`.

grammar	seed	n_layer	n_embd	n_head	steps	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	1	128	4	4000	0.974	0.816	0.136	0.374
0	1	1	128	4	4000	0.933	0.847	0.131	0.403
0	2	1	128	4	4000	1.028	0.776	0.114	0.354
1	0	1	128	4	4000	0.913	0.858	0.271	0.475
1	1	1	128	4	4000	0.873	0.887	0.268	0.483
1	2	1	128	4	4000	0.854	0.901	0.245	0.421
2	0	1	128	4	4000	0.92	0.875	0.292	-0.138
2	1	1	128	4	4000	0.872	0.912	0.248	0.344
2	2	1	128	4	4000	0.911	0.882	0.298	0.382
0	0	2	16	4	4000	1.065	0.749	0.022	0.172
0	1	2	16	4	4000	1.048	0.762	0.045	0.203
0	2	2	16	4	4000	1.038	0.769	0.032	0.183
1	0	2	16	4	4000	0.966	0.819	0.065	0.233
1	1	2	16	4	4000	0.951	0.83	0.079	0.216
1	2	2	16	4	4000	0.921	0.852	0.073	0.222
2	0	2	16	4	4000	0.943	0.857	0.097	-0.115
2	1	2	16	4	4000	0.973	0.835	0.08	0.096
2	2	2	16	4	4000	0.934	0.864	0.107	0.12
0	0	2	64	4	4000	1.107	0.718	0.143	0.329
0	1	2	64	4	4000	1.236	0.623	0.127	0.358
0	2	2	64	4	4000	1.225	0.631	0.151	0.351
1	0	2	64	4	4000	1.05	0.757	0.258	0.397
1	1	2	64	4	4000	1.182	0.66	0.225	0.381
1	2	2	64	4	4000	1.084	0.732	0.212	0.343
2	0	2	64	4	4000	1.182	0.677	0.249	-0.121
2	1	2	64	4	4000	1.122	0.722	0.283	0.304
2	2	2	64	4	4000	1.103	0.737	0.258	0.296
0	0	2	128	1	4000	0.919	0.857	0.205	0.458
0	1	2	128	1	4000	0.936	0.844	0.165	0.468
0	2	2	128	1	4000	1.07	0.745	0.155	0.432
1	0	2	128	1	4000	0.931	0.844	0.266	0.527

grammar	seed	n_layer	n_embd	n_head	steps	test_ce	loss_gap_closed	root_r2	deepest_r2
1	1	2	128	1	4000	0.943	0.836	0.274	0.512
1	2	2	128	1	4000	0.931	0.845	0.287	0.495
2	0	2	128	1	4000	0.916	0.878	0.369	0.091
2	1	2	128	1	4000	0.937	0.862	0.322	0.449
2	2	2	128	1	4000	1.02	0.799	0.247	0.371
0	0	2	128	2	4000	0.891	0.878	0.288	0.508
0	1	2	128	2	4000	0.871	0.893	0.269	0.594
0	2	2	128	2	4000	0.899	0.871	0.17	0.465
1	0	2	128	2	4000	0.856	0.9	0.449	0.585
1	1	2	128	2	4000	0.81	0.934	0.432	0.619
1	2	2	128	2	4000	0.867	0.892	0.344	0.495
2	0	2	128	2	4000	1.023	0.797	0.37	-0.19
2	1	2	128	2	4000	0.877	0.908	0.353	0.453
2	2	2	128	2	4000	0.859	0.921	0.365	0.477
0	0	2	128	4	4000	1.019	0.783	0.182	0.409
0	1	2	128	4	4000	0.964	0.824	0.214	0.498
0	2	2	128	4	4000	0.877	0.888	0.301	0.515
1	0	2	128	4	4000	0.89	0.875	0.467	0.576
1	1	2	128	4	4000	0.832	0.918	0.396	0.592
1	2	2	128	4	4000	0.819	0.927	0.405	0.54
2	0	2	128	4	4000	0.837	0.938	0.418	-0.028
2	1	2	128	4	4000	0.935	0.864	0.37	0.404
2	2	2	128	4	4000	0.883	0.903	0.426	0.464
0	0	2	128	4	16000	1.307	0.571	0.304	0.477
0	1	2	128	4	16000	1.17	0.671	0.298	0.566
0	2	2	128	4	16000	1.278	0.592	0.304	0.517
1	0	2	128	4	16000	1.086	0.73	0.406	0.558
1	1	2	128	4	16000	1.074	0.739	0.442	0.603
1	2	2	128	4	16000	0.972	0.814	0.474	0.583
2	0	2	128	4	16000	0.984	0.827	0.497	-0.171
2	1	2	128	4	16000	1.096	0.742	0.478	0.552
2	2	2	128	4	16000	0.895	0.894	0.448	0.506
0	0	2	128	8	4000	1.046	0.763	0.253	0.461
0	1	2	128	8	4000	1.016	0.786	0.272	0.532
0	2	2	128	8	4000	0.917	0.859	0.269	0.521
1	0	2	128	8	4000	0.936	0.841	0.456	0.558
1	1	2	128	8	4000	0.948	0.832	0.369	0.527
1	2	2	128	8	4000	0.82	0.926	0.369	0.488
2	0	2	128	8	4000	0.885	0.901	0.431	-0.051
2	1	2	128	8	4000	0.903	0.888	0.405	0.456
2	2	2	128	8	4000	0.896	0.893	0.439	0.488
0	0	2	256	4	4000	0.891	0.877	0.292	0.635
0	1	2	256	4	4000	0.887	0.88	0.21	0.546
0	2	2	256	4	4000	0.833	0.92	0.329	0.673
1	0	2	256	4	4000	0.86	0.897	0.518	0.693
1	1	2	256	4	4000	0.828	0.92	0.646	0.775
1	2	2	256	4	4000	0.803	0.939	0.627	0.753
2	0	2	256	4	4000	0.912	0.881	0.405	0.167
2	1	2	256	4	4000	0.895	0.894	0.344	0.489
2	2	2	256	4	4000	0.89	0.898	0.54	0.609
0	0	3	128	4	4000	0.937	0.844	0.297	0.514
0	1	3	128	4	4000	0.867	0.895	0.319	0.581
0	2	3	128	4	4000	0.883	0.883	0.3	0.513
1	0	3	128	4	4000	0.825	0.923	0.443	0.562
1	1	3	128	4	4000	0.882	0.881	0.398	0.548
1	2	3	128	4	4000	0.803	0.939	0.36	0.522
2	0	3	128	4	4000	0.855	0.924	0.414	-0.137
2	1	3	128	4	4000	0.861	0.92	0.47	0.506
2	2	3	128	4	4000	0.865	0.916	0.463	0.508
0	0	4	128	4	4000	0.884	0.883	0.311	0.525
0	1	4	128	4	4000	0.83	0.922	0.272	0.564
0	2	4	128	4	4000	0.836	0.918	0.288	0.557
1	0	4	128	4	4000	0.803	0.939	0.502	0.602
1	1	4	128	4	4000	0.838	0.913	0.461	0.61

grammar	seed	n_layer	n_embd	n_head	steps	test_ce	loss_gap_closed	root_r2	deepest_r2
1	2	4	128	4	4000	0.804	0.938	0.442	0.529
2	0	4	128	4	4000	0.853	0.925	0.46	-0.082
2	1	4	128	4	4000	0.832	0.941	0.5	0.537
2	2	4	128	4	4000	0.83	0.943	0.497	0.508

Table 4: All 99 architecture-sweep runs (11 one-axis-at-a-time capacity configs × 3 grammars × 3 seeds). No convergence filtering; small models are expected to underfit.

12. Appendix: Per-run $L = 4$ depth-sweep sanity table

Every run in the $L = 4$ depth sweep, no filtering — all 60 (10 grammars × 3 seeds × $n_{\text{layer}} \in \{2, 3\}$). `grammar` and `seed` index the rule-table draw and master replicate seed; `n_layer` is the only varied architecture axis ($n_{\text{embd}} = 128$, $n_{\text{head}} = 4$, 4000 steps throughout). `test_ce` is the held-out next-token cross-entropy; `loss_gap_closed` the fraction of the uniform→Bayes gap closed (covariate, not a gate); `root_r2` and `deepest_r2` the level-L_0 and mean leaf-parent-L_3 probe R^2. Loaded directly from `figures/depth4_sanity.csv`.

grammar	seed	n_layer	test_ce	loss_gap_closed	root_r2	deepest_r2
0	0	2	0.716	0.992	0.207	0.32
0	0	3	0.716	0.991	0.23	0.35
0	1	2	0.715	0.992	0.178	0.345
0	1	3	0.719	0.99	0.263	0.386
0	2	2	0.728	0.983	0.148	0.307
0	2	3	0.711	0.995	0.276	0.33
1	0	2	0.728	0.982	0.227	0.284
1	0	3	0.713	0.993	0.282	0.298
1	1	2	0.715	0.992	0.227	0.28
1	1	3	0.718	0.989	0.288	0.341
1	2	2	0.722	0.987	0.179	0.273
1	2	3	0.715	0.992	0.298	0.336
2	0	2	0.724	0.987	0.203	0.317
2	0	3	0.726	0.986	0.271	0.372
2	1	2	0.727	0.985	0.171	0.304
2	1	3	0.718	0.992	0.301	0.384
2	2	2	0.725	0.986	0.149	0.315
2	2	3	0.72	0.99	0.318	0.386
3	0	2	0.74	0.986	0.231	0.25
3	0	3	0.732	0.992	0.304	0.235
3	1	2	0.783	0.955	0.22	0.257
3	1	3	0.73	0.994	0.351	0.244
3	2	2	0.75	0.979	0.211	0.273
3	2	3	0.735	0.99	0.345	0.301
4	0	2	0.73	0.987	0.183	0.381
4	0	3	0.722	0.992	0.33	0.423
4	1	2	0.723	0.992	0.171	0.347
4	1	3	0.731	0.986	0.2	0.368
4	2	2	0.722	0.992	0.244	0.388
4	2	3	0.721	0.993	0.308	0.443
5	0	2	0.745	0.978	0.073	0.273
5	0	3	0.732	0.988	0.134	0.288
5	1	2	0.741	0.981	0.103	0.317
5	1	3	0.728	0.991	0.205	0.33
5	2	2	0.748	0.976	0.134	0.283
5	2	3	0.743	0.98	0.153	0.306
6	0	2	0.718	0.992	0.213	0.31
6	0	3	0.72	0.991	0.29	0.386
6	1	2	0.731	0.983	0.134	0.29
6	1	3	0.731	0.982	0.206	0.351
6	2	2	0.723	0.988	0.138	0.299
6	2	3	0.717	0.993	0.266	0.34
7	0	2	0.732	0.987	0.254	0.259
7	0	3	0.744	0.978	0.228	0.274
7	1	2	0.729	0.989	0.227	0.103

grammar	seed	n_layer	test_ce	loss_gap_closed	root_r2	deepest_r2
7	1	3	0.762	0.965	0.249	0.121
7	2	2	0.732	0.987	0.186	-0.229
7	2	3	0.737	0.983	0.245	-0.216
8	0	2	0.718	0.988	0.202	0.292
8	0	3	0.714	0.99	0.257	0.33
8	1	2	0.727	0.981	0.236	0.276
8	1	3	0.713	0.991	0.294	0.315
8	2	2	0.732	0.978	0.152	0.245
8	2	3	0.718	0.988	0.327	0.347
9	0	2	0.727	0.986	0.27	0.298
9	0	3	0.727	0.986	0.383	0.348
9	1	2	0.721	0.991	0.264	0.328
9	1	3	0.716	0.994	0.379	0.387
9	2	2	0.717	0.993	0.298	0.326
9	2	3	0.724	0.988	0.333	0.351

Table 5: All 60 $L = 4$ depth-sweep runs (10 grammars × 3 seeds × $n_{\text{layer}} \in \{2, 3\}$). No convergence filtering. `deepest_r2` is the mean over the eight leaf-parent (L_3) nodes.

13. Appendix: Per-run non-collapse sweep sanity table

Every run in the non-collapse sweep, no filtering — all 27 (3 `skew` × 3 `ambiguity` × 3 rule-table draws), at the pinned arch ($n_{\text{layer}} = 2, n_{\text{embd}} = 128, n_{\text{head}} = 4$, 4000 steps). `bayes_floor` is the exact weighted Bayes-optimal next-token entropy; `loss_gap_closed` the fraction of the uniform→Bayes gap closed (covariate); `root_r2`/`mid_r2`/`low_r2` the level-$L_0/L_1/L_2$ probe R^2 (shuffled baseline `shuffled_r2`); `k8_entropy` the headline mean $k = 8$ root posterior entropy; `eff_dim`/`hull_area` the reachable-set descriptors. Loaded directly from `figures/noncollapse_sanity.csv`.

skew	ambiguity	draw	test_ce	bayes_floor	loss_gap_closed	root_r2	mid_r2	low_r2	shuffled_r2	k8_entropy	eff_dim	hull_area
high	0.0000	0	1.1915	0.2818	0.4939	0.2369	0.3522	0.4214	-0.0868	0.0000	6.9652	0.6465
high	0.0000	1	0.8751	0.3953	0.7151	0.2492	0.3554	0.4202	-0.0628	0.0000	6.9652	0.6465
high	0.0000	2	1.0670	0.2532	0.5544	0.2983	0.1920	0.3542	-0.0782	0.0000	6.9652	0.6465
high	0.3000	0	0.9041	0.2118	0.6293	0.3115	0.3577	0.5997	-0.0447	0.2655	5.6327	0.5984
high	0.3000	1	0.3135	0.1622	0.9211	0.4511	0.5554	0.5639	-0.0808	0.3314	4.9094	0.6720
high	0.3000	2	0.6246	0.2468	0.7938	0.3783	0.4539	0.5344	-0.1115	0.4441	4.7952	0.6780
high	0.6000	0	0.3267	0.1826	0.9241	0.5202	0.6696	0.6840	-0.0675	0.5852	3.7205	0.6501
high	0.6000	1	0.2625	0.2101	0.9720	0.5496	0.6274	0.8281	-0.0704	0.6123	3.6509	0.7863
high	0.6000	2	0.2848	0.2101	0.9601	0.5612	0.7059	0.7040	-0.1028	0.4243	2.8199	0.6248
mid	0.0000	0	0.8921	0.5366	0.7696	0.3287	0.4008	0.3964	-0.0608	0.0000	6.9652	0.6465
mid	0.0000	1	1.0199	0.5468	0.6913	0.3799	0.5235	0.5137	-0.0439	0.0000	6.9652	0.6465
mid	0.0000	2	1.1025	0.5426	0.6357	0.3549	0.3549	-7.4713	-0.0763	0.0000	6.9652	0.6465
mid	0.3000	0	0.6272	0.4867	0.9118	0.4159	0.4348	0.5595	-0.0552	0.6537	4.8768	0.5921
mid	0.3000	1	0.6823	0.4995	0.8843	0.4557	0.6463	0.6656	-0.0553	0.3870	5.8288	0.6014
mid	0.3000	2	0.6325	0.5326	0.9354	0.3626	0.4213	0.6025	-0.0270	0.7169	5.2545	0.7321
mid	0.6000	0	0.5376	0.4261	0.9326	0.5044	0.6875	0.8028	-0.0715	1.1605	4.3417	0.5354
mid	0.6000	1	0.6313	0.5088	0.9220	0.3874	0.5424	0.6770	-0.0920	0.9423	3.8371	0.6337
mid	0.6000	2	0.5529	0.4420	0.9323	0.4711	0.7290	0.7901	-0.1003	1.0470	4.0308	0.4432
none	0.0000	0	0.8900	0.7253	0.8784	0.3002	0.4092	0.5039	-0.0568	0.0000	6.9652	0.6465
none	0.0000	1	0.8494	0.7199	0.9047	0.3578	0.5082	0.5504	-0.0638	0.0000	6.9652	0.6465
none	0.0000	2	0.8447	0.7544	0.9319	0.4262	0.4582	0.0083	-0.0640	0.0000	6.9652	0.6465
none	0.3000	0	0.6930	0.6221	0.9513	0.4455	0.4894	0.6512	-0.0439	0.7598	5.3126	0.6421
none	0.3000	1	0.7216	0.6509	0.9505	0.4274	0.6678	0.6417	-0.0553	0.6576	5.5352	0.5416
none	0.3000	2	0.7397	0.6660	0.9479	0.3771	0.5106	0.5869	-0.0733	0.7003	5.4769	0.5762
none	0.6000	0	0.5011	0.4860	0.9905	0.5640	0.6747	0.7615	-0.1606	1.3969	3.7241	0.6400
none	0.6000	1	0.5658	0.5448	0.9863	0.6170	0.7852	0.9018	-0.0745	1.3457	3.3016	0.5520
none	0.6000	2	0.5328	0.4976	0.9778	0.4220	0.6758	0.7826	-0.0481	1.4068	2.7883	0.4318

Table 6: All 27 non-collapse runs (3 skew × 3 ambiguity × 3 draws). No filtering. `k8_entropy` ≈ 0 exactly when $\rho = 0$ regardless of skew.

14. References

Cagnetta, F., Favero, A., Sclocchi, A., and Wyart, M. (2025). **Scaling Laws and Representation Learning in Simple Hierarchical Languages: Transformers vs. Convolutional Architectures**. arXiv:2505.07070.

Shai, A. et al. (2026). **Transformers learn factored representations**. arXiv:2602.02385.

Project sources: `spec.md`, `spec-grammar-sweep.md`, `PREREGISTRATION.md`, `EXECUTION_OUTPUT.md`, `results/analysis.json`, `results/root_reconstruction.json`, `artifacts/train_summary.json`. Grammar sweep: `sweep.py`,

`sweep.yaml`, `results/sweep/g*/{config.json,analysis.json}`, `figures/sweep_sanity.csv`. Architecture sweep: `spec-arch-sweep.md`, `run_arch.py`, `results/arch/g*/{config.json,analysis.json}`, `figures/arch_sanity.csv`.